PART

Eight

References

CHAPTER 39

Selected Global Bond Markets

The following section provides a short overview of the main global government bond markets. Individual bond series are given using their usual abbreviation (ticker symbol). Where Bloomberg uses different tickers, they are given as well. Local language names for each series are provided where feasible. The list is deliberately not trying to be exhaustive and some details are bound to change over time. Some additional markets are provided for illustration.

39.1 EURO AREA

The Euro area comprises the countries that introduced the common currency, the euro. Currently, these countries are: Austria, Belgium, Cyprus, Estonia, Finland, France, Germany, Greece, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, The Netherlands, Portugal, Slovakia, Slovenia and Spain. The euro was introduced in 1999 and other countries joined later.

Each country in the euro area had its own debt market before the euro was introduced, and with the exception of the German market, where the role of the Deutsche Mark as a reserve currency had already brought in large foreign participation, most of these markets were largely (up to 95%) domestic. In order to satisfy the demands of the domestic client base, bond issues tended to be frequent and small because the low amount of secondary market trading made liquidity concerns unimportant. Bank trading desks tended to have specialised traders for each market. With the euro came a greater focus on cross-market trading, liquidity, and foreign investor participation. Two processes were initiated by the sovereigns in this context, and these were redenomination and reconventioning.

Redenomination refers to stating cash flows (coupons, redemption amound, and prices) in terms of euros instead of the legacy currencies. This was a largely symbolic step because the laws introducing the euro in the member countries retained the legal force of contracts stated in terms of the original currencies and effectively brought about redenomination anyway[1]. Reconventioning, on the other hand, was the very important

[1]Legally the legacy currencies still exist but are now non-decimal subunits of the euro. Saying 10,000 lire is now just an unusual way of saying 5.16 euro. The reason redenomination was undertaken is that exact instructions are required for the clearing of transactions. Computer systems

decision to make the bond markets more uniform by changing all daycount conventions to act/act and abolish minimum denominations. Given that the legacy markets used all kinds of conventions from 30/360 to act/act, this greatly aided the transparency of the market. Reconventioning was generally applied as of the first coupon payment date after the introduction of the euro, so each bond in theory was reconventioned on a different day. Note however, that there are still subtle differences in place between the markets, such as accrued interest rounding conventions in Italy. However, bond markets are clearly becoming more integrated and from around 2001 banks have started to reorganise their trading desks more along maturity-based splits than on issuer lines.

With the greater stability in short term interest rates brought about by the larger currency area, sovereigns have generally decided to lower the average time to maturity of their outstanding debt. Note that large issuers, such as sovereigns, have to make a conscious trade-off between long-term stability of financing costs (represented by long maturity bonds) and lower current interest rates (such as afforded by short maturity bonds). There is no hard and fast way of estimating the optimal maturity profile, but generally the ideal average time to maturity falls with decreasing rate volatility. By today, practically all sovereigns have started to use interest rate swaps as a tool to manage duration risk, but the sheer size of sovereign debt and the fact that practically all sovereigns are trying to manage their debt duration in the same direction, mean that there is limited scope for the market to absorb the risks. The French Trésor announced a large swap programme in 2001 only to find that the market almost immediately repriced the swap spreads of French government bonds.

With the exception of the German government, euro area sovereigns do not commit themselves to issue a certain amount of debt of a particular maturity far ahead of time. This gives them the flexibility to adjust the size, if not the maturity, of the issued bonds to the market environment at the time of the auction or syndication. Such flexibility can be both a good and a bad thing. The benchmark status of the German government debt is probably helped by the fact that the German agency does not attempt to time the market too much. However, only one issuer can be the benchmark issuer and, for the other governments, flexibility may therefore be more important. That being said, the bulk of issuance can be forecast with reasonable accuracy in most countries.

Supply in the euro area is generally through auction for the bigger issuers and syndication for the smaller ones. The EuroMTS interdealer trading system imposes a minimum size of EUR 5bn for bonds to be listed. This provides a strong incentive for debt managers to provide this amount of debt with the initial tranche of a new bond and restricts smaller sovereigns in terms of the number of bonds they can issue per year. Such smaller issuers will usually place the large initial tranche via syndication and then conduct regular tap auctions.

39.1.1 Austria

Austrian government debt is managed by the Austrian Federal Financing Agency (Österreichische Bundesfinanzierungsagentur, www.oebfa.co.at). Austrian government

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are not given to analysing the economics of a transaction to decide whether instructions given in different currencies amount to the same economic value.

bonds used to be an example of having many small issues at very similar maturities, but quickly switched to issuing fewer, but larger benchmark bonds after the introduction of the euro.

RAGB (Bundesanleihe)	
Maturities	10Y, 15Y, 30Y, 50Y, 70Y, 100Y
Coupon	Fixed annual, both decimal and 1/4% steps
Yield convention	True Street (annual compound)
Settlement, daycount	T+2 Vienna, act/act
Issuance	Monthly auctions or syndication
Usual volume	10bn
Strippable	Yes

RATB (Bundesschatzschein)	
Maturities	7–365 days
Coupon	Zero coupon
Yield convention	Simple yield
Settlement, daycount	T+2, depending on market
Issuance	Irregular
Usual volume	Irregular
Strippable	No

Remarks:

Austrian T-bills are usually issued under English law even when denominated in euros. There is also no cross-default clause.

39.1.2 Belgium

Belgian government debt is managed by the Belgian State Treasury (treasury.fgov.be) which is part of the Ministry of Finance of the Kingdom of Belgium. The Treasury was an early adopter of exchange auctions with a view to consolidate illiquid product lines. In contrast to most other treasuries, Belgium also actively used exchange auctions to smoothen out the redemption profile of State debt. To this end, bonds within 1 year of maturity were exchanged against current long-dated benchmark bonds.

OLO/BGB (Obligations linéaires/Lineaire obligaties)	
Maturities	2Y, 5Y, 10Y, 30Y, 50Y
Coupon	Fixed annual, 1/4% steps, now decimal
Yield convention	True Street, annual compound
Settlement, daycount	T+2 Brussels, act/act
Issuance	Monthly auction or syndication
Usual volume	7.5–15bn
Strippable	Yes

BGTB (Certificats de Trésorerie/Schatkistcertificaten)	
Maturities	3M, 6M, 12M
Coupon	Zero coupon
Yield convention	Simple yield
Settlement, daycount	T+2 Brussels, act/360
Issuance	Biweekly auction
Usual volume	1.5–2bn
Strippable	No

Remarks:

Belgium was the only euro area money market that had to switch from act/365 to act/360 daycount convention. Note that Luxembourg had a currency union with Belgium before adopting the euro.

39.1.3 Finland

Finnish debt is managed by the State Treasury (www.valtiokonttori.fi) and has an EMTN programme that allows it to place public debt at short notice.

RFGB (Sarjaobligaation)	
Maturities	5Y, 10Y
Coupon	Fixed annual, 1/8% steps
Yield convention	True Street, annual compound
Settlement, daycount	T+2 Helsinki, act/act
Issuance	Usually syndication followed by tap auctions. Although an auction calendar exists, scheduled auctions dates are frequently skipped.
Usual volume	5–7bn
Strippable	No

39.1.4 France

French government debt is issued by the Agence France Trésor (www.aft.gouv.fr), an agency of the French State set up in February 2001. AFT is active in the swap market and also conducts sporadic buy-backs of older French bonds. The issuance policy of the Tresor combines a fixed calendar with some flexibility as to the supply amount and the bonds supplied. France has in the past tapped off-the-run bonds which is otherwise fairly uncommon in any market.

France conducts two auctions per month, with a 10Y or longer OAT auction every first Thursday of every month and a 2Y, 5Y and inflation-linked OATi/OAT€i auction each third Thursday. These auction dates can be shifted if they coincide with national holidays. France has a primary dealer system called SVT (Spécialistes en Valeurs du Trésor) and a lending facility where SVT can borrow bonds to cover short positions.

OAT/FRTR (Obligations Assimilables du Trésor)

Maturities	10Y, 15Y, 30Y, 50Y
Coupon	Fixed annual, 1/4% steps
Yield convention	True Street (annual compound)
Settlement, daycount	T+2 Paris, act/act
Issuance	Usually two new 10Y issues per year, one 30Y, occasionally 15Y, and first 50Y in 2005
Usual volume	10–45bn
Strippable	Yes, coupon and principal strips are fungible

Remarks:

The Trésor is one of the more innovative global debt managers and the variety of OAT issues is evidence of that. France had issued Ecu-denominated OATs that were funged with French Franc OATs into euro issues, there were TEC10 10Y floating rate linked OATs, France issued the first 50Y government bond in euros, and so on. For a while, there was some effort to wrest benchmark status from the German 10Y sector through support of the Matif 10Y contract, but this failed due to a lack of market interest. The French curve remains an important reference or euro-area sub-sovereign issuers, in part due to the proliferation of French agency issuers.

OATi and OAT€i (Obligations Assimilables du Trésor indexées)

Maturities	10Y, 30Y
Coupon	Fixed annual, decimal 0.05% steps, linked to French CPI ex-tobacco (OATi) or Euroland HICP ex-tobacco (OAT€i)
Yield convention	True Street (annual compound), inflation-linked
Settlement, daycount	T+2 Paris, act/act
Issuance	Irregular
Usual volume	Irregular
Strippable	No

Remarks:

The French government was the first one to issue inflation-linked debt in the euro area and remains the benchmark issuer in this regard. Originally, the inflation link was to French CPI excluding tobacco costs, but bonds linked to euro area HICP inflation were first issued in 2001 and proved rather popular. Instead of switching completely to issuance of OAT€i, the Trésor still issues OATi and allocates supply opportunistically between the two.

BTAN/BTNS (Bons du Trésor à taux fixe et intérèts annuels)

Maturities	2Y, 5Y
Coupon	Fixed annual, 1/4% steps
Yield convention	True Street (annual compound)
Settlement, daycount	T+2 Paris, act/act
Issuance	Usually two 2Y and 5Y issues per year
Usual volume	20bn
Strippable	No

Remarks:

BTNS for a while held the 5Y benchmark status in euros and were commonly used to price new corporate supply. Quotation and settlement convention for BTNS differed from OAT but were gradually brought in line. France switched completely from issuing BTNS to FRTR/OAT.

BTF (Bons du Trésor à taux fixe et à intérèt précompté)

Maturities	13, 26, 52 weeks
Coupon	Zero coupon
Yield convention	simple yield
Settlement, daycount	T+1 Paris, act/360
Issuance	Weekly auction
Usual volume	4bn
Strippable	No

Remarks:

BTFs form the most liquid bill market in euros.

39.1.5 Germany

German government debt used to be managed by the German Bundesbank, but in 2001 the German Ministry of Finance set up a private company wholly owned by the German State to manage the debt instead. This agency is called the German Financing Agency (Deutsche Finanzagentur, www.deutsche-finanzagentur.de) and is located in Frankfurt.

The German Agency, like the Bundesbank before it, is very active in the secondary market (both outright and repo) through a process called market management, and retains part of each auction for this purpose. Market management is done mainly through anonymous interdealer platforms, so there is no way of gauging the Agency's actions. The Agency is also a fairly active user of interest rate swaps. Traditionally, Germany saw steady demand for its debt because the Deutsche Mark used to be a reserve currency, the German credit was perceived as very good, and supply was reasonably limited. With the introduction of the euro and the high cost imposed by the German reunification process, this uniquely benevolent situation is no longer present and the Agency is active in promoting German debt.

German bond auctions always happen on Wednesdays with the exception of Bubills which are issued on Mondays and inflation-linked bonds on Tuesdays. BKO auctions switched to Tuesdays in 2017. There is no primary dealer systems and bidding at the auctions is done by an auction bidding group that is in principle open to all dealers. Bidders are ejected from this group if they fail to take down at least 0.1% of the annual supply in duration-ajusted terms. The Agency publishes an annual issuance outlook from which it is generally possible to infer auction dates and auction amounts for the whole year with only marginal uncertainty. Towards the each quarter, the Agency publishes a detailed calendar for the following quarter. Deviations from the auction calendar (usually extra supply) can take place via non-public taps into the market management account of the Agency. Although they are announced in a timely fashion, it is impossible to gauge the market impact of such taps because it is not clear how long the Agency will retain these bonds.

Because the Agency can withold an arbitrary amount of debt at any auction, there is never a need to lower the amount issued at an auction. Because Germany typically tends to receive a large amount of noncompetitive bids and has complete freedom in allocating to these bids, it is not trivial to gauge the actual demand situation at an auction from the observed prices.

DBR (Bundesanleihe)	
Maturities	10Y, 30Y, exceptionally 15Y
Coupon	Fixed annual, usually 1/4% steps
Yield convention	True Street (annual compound)
Settlement, daycount	T+2 Frankfurt, act/act
Issuance	Two 10Y issues per year, 0–1 30Y per year, typically 2–3 taps for each new 10Y
Usual volume	around 25bn
Strippable	Yes

Remarks:

DBR issues are the main benchmarks for European yields. This partly has historical reasons (some central banks still have restrictions for investing in other euro area sovereigns whereas the Deutsche Mark used to be a reserve currency), but is also supported by the dominance of the Eurex bond futures contracts in euro area trading. The world's most active contract, the Eurex Bund, is a contract on 10Y DBR, and the 5Y Eurex Bobl contract usually has a DBR CTD every other delivery. DBR issues tend to have long first coupons.

OBL (Bundesobligation)

Maturities	5Y
Coupon	fixed annual, usually 1/4% steps
Yield convention	True Street (annual compound)
Settlement, daycount	T+2 Frankfurt, act/act
Issuance	Two issues per year, typically two taps for each new issue
Usual volume	20bn
Strippable	No

Remarks:

The issuance procedures and maturity months for OBLs have changed several times since the introduction of the euro because these bonds were also important retail instruments. The current system is fairly transparent and fully adapted to the requirements of the wholesale market. The supply is now timed so that every other 5Y contract (Bobl) on Eurex will have an OBL as CTD.

DBRi (Inflationsindizierte Bundesanleihe)

Maturities	10Y, 30Y
Coupon	Fixed annual, decimal, linked to Euroland HICP ex-tobacco
Yield convention	True Street (annual compound)
Settlement, daycount	T+2 Frankfurt, act/act
Issuance	Irregular
Usual volume	10–16bn
Strippable	No

Remarks:

Inflation-linked contracts were severely restricted in Germany by the law that introduced the Deutsche Mark because such obligations were seen as inflationary feedback loops. This restrction was removed with the introduction of the euro, but the government did not start to issue inflation-linked bonds until 2006. For some time, there were also indexed 5Y OBLi. German break-even inflation trades very close to French break-even (when adjusted for seasonality) and the French inflation-linked curve remains the benchmark.

BKO (Bundesschatzanweisung, formerly Bundeskassenobligation)	
Maturities	2Y
Coupon	Fixed annual, usually 1/4% steps
Yield convention	True Street (annual compound)
Settlement, daycount	T+2 Frankfurt, act/act
Issuance	Four issues per year, March, June, September, and December maturities
Usual volume	12bn
Strippable	No

Remarks:

Eurex Schatz contracts tend to have a BKO CTD and because the maturities tend to fall in the middle of the month, BKO swap spreads can be traded using Euribor contracts with little basis risk.

BUBILL (Bundeskassenschein)	
Maturities	6M
Coupon	Zero-coupon
Yield convention	Simple yield
Settlement, daycount	T+2 Frankfurt, act/360
Issuance	Twelve issues per year
Usual volume	6bn
Strippable	No

Remarks:

The German government had a tacit agreement with the Bundesbank not to be overly active in the short-term debt market to avoid interference with the money supply targets of the central bank. With the introduction of the euro, the situation changed and Bubills became more active. However, their liquidity remains inferior to the French BTFs.

39.1.6 Greece

Due to problems with achieving the strict debt criteria needed to qualify, Greece joined the euro only in 2001, just in time for the introduction of euro cash. Due to excessive debts, Greece in 2010 needed a rescue package from the other EU member states who set up the EFSM and EFSF for this purpose. Greece restructured the bulk of its outstanding debt in 2012 as part of this rescue effort and remains sub-investment grade. The support for Greece was redesigned several times to achieve debt sustainability, and Greece has regained market access. The first post-restructuring bonds were issued already in 2014. The Greek bonds issued while Greece was receiving fiscal support from other euro member states had higher coupons than the interest charged by the public creditors. Somewhat uniquely, therefore, creditors willing to ease payment terms receive a lower interest rate than creditors with fixed claims.

Greek government debt is managed by the Public Debt Management Agency (www.pdma.gr). The market is essentially split between the 'strip', which is a series of bonds allocated to investors who participated in the 2012 'private sector involvement' (PSI), and the post-PSI bonds. The liquidity of the strip is poor while the newer bonds trade somewhat more actively.

GGB (ομολογο ελληνικου δημοσιου)	
Maturities	3Y, 5Y, 10Y
Coupon	Fixed annual, 1/8% steps
Yield convention	True street, annual compound
Settlement, daycount	T+2 Athens, ISMA 30/360 or act/act
Issuance	Monthly auctions
Usual volume	5–10bn
Strippable	No

Remarks:
Because the Bank of Greece operates a securities depository, transaction volumes of Greek government bonds are quite transparent.

39.1.7 Ireland

Irish debt is managed by the National Treasury Management Agency NTMA (www.ntma.ie). Through a large bond exchange auction and listing in Euroclear in 2000, Ireland brought more liquidity into the market, and dispersion of liquidity across different instruments, such as CP, has also been reduced. Ireland issues both domestic euro debt and foreign currency bonds and also maintains an active CP programme. Ireland suffered heavily in the financial crisis due to an outsized financial system.

IRISH	
Maturities	2Y, 10Y, 15Y
Coupon	Fixed annual, decimal steps
Yield convention	True street (annual compound)
Settlement, daycount	T+2 Dublin, act/act
Issuance	Monthly auction (every third Thursday) or syndication
Usual volume	10bn
Strippable	No

39.1.8 Italy

The Italian Treasury (www.dt.tesoro.it) is part of the Ministry of Finance. Given the large size of Italy's debt relative to its GDP, Italy has been innovative in its debt management policy and has been one of the earliest adopters of derivatives. Italy has a highly liquid market in Treasury bills and when the floating rate CCTs are added to the actual zero coupon instruments, a large share of the total Italian government debt is short-dated.

BTPS (Buoni poliennali del Tesoro)	
Maturities	3Y, 5Y, 10Y, 15Y, 30Y
Coupon	Fixed semi-annual, 1/4% steps and decimal
Yield convention	True Street, bond-equivalent
Settlement, daycount	T+2 Milan, act/act
Issuance	Two auctions per month, mid-month for for 5Y, 15Y and 30Y, month end for 3Y and 10Y
Usual volume	10–30bn
Strippable	Yes

Remarks:
The trading platform for BPT, MTS, is now the dominant interdealer platform in the euro area.

CCT/CCTS (Certificati di Credito del Tesoro)	
Maturities	7Y
Coupon	Floating rate semi-annual, linked to most recent BOT auction yield plus margin
Yield convention	Not applicable
Settlement, daycount	T+2 Milan
Issuance	discontinued
Usual volume	not applicable
Strippable	No

Remarks:
CCT have been discontinued due to low liquidity and some questionable market practices. CCT used to be a conventient instrument for savings banks because both the variable coupon, and the 7Y maturity matches well the characteristics of a consumer deposit. Their role has been taken over by CCT€.

CCT€/CCTS (Certificati di Credito del Tesoro €)	
Maturities	7Y
Coupon	Floating rate semi-annual, linked to Euribor plus margin
Yield convention	Not applicable
Settlement, daycount	T+2 Milan
Issuance	Monthly auction at month-end (together with 3Y and 10Y BTPS)
Usual volume	12–18bn
Strippable	No

Remarks:
CCT€ avoid some of the problems connected to the BOTS auction link of CCT but remain less liquid than BTPs.

CTZ/ICTZ (Certificati del Tesoro Zero Coupon)	
Maturities	2Y
Coupon	Zero coupon
Yield convention	Simple yield
Settlement, daycount	T+2 Milan, act/365 (note: not act/360)
Issuance	Monthly auction
Usual volume	15–18 bn
Strippable	No

Remarks:
CTZ are exceptional in that they are included in bond indices despite being issued as bills.

BOT/BOTS (Buoni Ordinari del Tesoro)	
Maturities	3, 6, 12 months
Coupon	Zero coupon
Yield convention	Simple yield
Settlement, daycount	T+2 Milan, act/360
Issuance	Monthly auction
Usual volume	6–10 bn
Strippable	No

39.1.9 The Netherlands

The debt of the Kingdom of the Netherlands is managed by the Dutch State Treasury Agency (www.dsta.nl). The Agency uses an auction method system called Dutch Direct Auction which is unique to The Netherlands. Auctions are done using an auction 'window' where the Agency continuously displays an offer price at which primary dealers can buy bonds[2]. The price is adjusted to dealer demand and market levels for up to one hour or until the desired bond volume has been sold. In such an auction, investment banks direct clients towards the DSTA. This is a mix between syndication (in that marketing is done by investment banks) and auction.

DSL/NETHER (Staatsleningen)	
Maturities	3Y, 10Y, 30Y
Coupon	Fixed annual, 1/4% steps
Yield convention	True Street, annual compound
Settlement, daycount	T+2 Amsterdam, act/act
Issuance	Monthly auction on the Tuesday following the first Wednesday
Usual volume	10–18bn
Strippable	Yes

Remarks:
DSTA offers a repo facility for bonds with issue sizes smaller than 5bn to market makers in order to avoid squeezes in new bonds.

DTC/DTB (Schatkistpapier)	
Maturities	3, 6, 9, 12M
Coupon	Zero coupon
Yield convention	Simple yield
Settlement, daycount	T+2 Amsterdam, act/360
Issuance	First Monday of each month
Usual volume	Irregular
Strippable	No

39.1.10 Portugal

The debt of the Republic of Portugal is managed by the Instituto de Gestão do Crédito Público (ICDO, www.igcp.pt). The Portuguese market remained fairly domestic even after the introduction of the euro because foreign investors could not clear government debt securities through either of the two major clearers Clearstream or Euroclear. Trading though electronic interdealer platforms also only started in earnest with the introduction of the Medip system which is operated by MTS Portugal. Nowadays, Portuguese government bonds can be cleared through Euroclear and the primary dealer group, Operadores Especializados en Valores do Tesouro, is fairly active.

[2]The expression refers to a time window.

PGB (Obrigações do Tesoro)	
Maturities	10Y, 30Y
Coupon	Fixed annual, 1/8% and decimal steps
Yield convention	True Street, annual compound
Settlement, daycount	T+2 Lisboa
Issuance	Syndication and monthly auction each second Wednesday
Usual volume	5–12bn
Strippable	No

39.1.11 Spain

The debt of the Kindom of Spain is managed by the Tesoro Público (www.tesoro.es), part of the Ministry of Finance. Spain has a primary dealer system. Eurex launched a Spanish government bond future in 2015 but the contract remains illiquid, partly due to competing with the Eurex 10Y and 3Y BTP contracts as a hedge instrument for spread products in euros, and partly due to the stability of the Spain–France spread.

SPGB (Bonos/Obligaciones)	
Maturities	3Y, 5Y (Bonos); 10Y, 15Y, 30Y (Obligaciones)
Coupon	Fixed annual, decimal 0.05% steps
Yield convention	True Street, annual compound
Settlement, daycount	T+2 Madrid, act/act
Issuance	Monthly except August, Bonos each first Thursday, Obligaciones each third Thursday
Usual volume	10bn
Strippable	Yes

Remarks:

Most Spanish Obligaciones used to have an unusual initial zero coupon period. The bonds were issued with less than one year to what would normally be the first coupon date but did not accrue any coupon during that period. Although the first coupon is therefore paid between one and two years after the bond is first issued, it is simply a full coupon, not a long first coupon. Put differently, the first accrual date is after the first settlement date. The bonds were therefore issued at a discount to the par price which effectively reflected negative accrued interest. This practice ended in 2006.

SPTB (Letras)	
Maturities	3, 6, 12, 18 months
Coupon	Zero coupon
Yield convention	Simple yield
Settlement, daycount	T+2 Madrid, act/360
Issuance	Monthly auction
Usual volume	5–8bn
Strippable	No
Remarks:	
The Spanish Letras are fairly liquid compared to the liquidity of Spanish coupon bonds.	

39.2 ICELAND

The Icelandic debt market is unique in that practically all long-term debt is inflation-linked, while all short-term debt is issued in nominal terms. This makes the analysis of break-even inflation practically impossible. This structure is the result of being a small open economy which makes FX rates an important driver of domestic inflation. All trades by market makers have to be reported to the Nasdaq Iceland (www.nasdaqomxnordic.com, formerly Icelandic Stock Exchange) and there is therefore a reasonable degree of price transparency.

The national debt of the Republic of Iceland is managed by the National Debt Management Agency (Lánamál Ríkisins, www.lanamal.is). The actual government debt sector is only a small part of the total debt market while the biggest issuer in the Icelandic market is the government Housing Financing Fund Íbúðalánasjórður (www.ils.is). Icelandic Housing bonds form the core of the Icelandic long-term debt market. The bonds are annuities linked to Icelandic inflation, which reflects the structure of the underlying mortgage loans. The debt of the Housing Financing Fund carries a deficiency guarantee of the Icelandic State but is not explicitly state-guaranteed.

RIKS/ICEGB (Ríkisbréf)	
Maturities	5Y, 10Y
Coupon	Annual fixed
Yield convention	True Street, compound
Settlement, daycount	T+2 Reykjavik, act/act
Issuance	Irregular auction
Usual volume	
Strippable	No

HFF/ICEHB (Íbúðalánasjórður Íbúðabréf)	
Maturities	2014, 2024, 2034, and 2044 maturities
Coupon	Fixed semi-annual 3.75%, amortising annuity, inflation-linked
Yield convention	True Street, bond-equivalent (international), semi-annual compound (domestic)
Settlement, daycount	T+1 Reykjavik, 30/360
Issuance	Irregular auction, private placements
Usual volume	
Strippable	No
Remarks:	
The current benchmark series of Icelandic Housing bonds was created in an exchange against older Húsbréf and Íbúðabréf bonds in 2004.	

39.3 JAPAN

Japan has the highest debt load of all OECD countries and unless drastic measures will be adopted, this debt load is likely to rise further as a result of demographic trends. Unlike other developed countries, Japan never conducted a currency reform to hide the inflationary effects of past wars and therefore the smallest currency denomination, the sen, is now only used to quote securities prices. Japan introduced a primary dealer system in 2004 and at the time of writing, there are 21 primary dealers.

Although the Japanese Yen is a central bank reserve currency and Japanese government debt represents a large part of global bond indices, Japanese bonds tend to be underrepresented in global investor portfolios. This is mainly due to the combination of low yields with a central bank policy to keep the Japanese Yen weak against the US dollar.

Japanese government bonds have an ex-coupon period of three days, but no settlement of the relevant bonds occurs during these three days. Instead, settlement that would fall into the ex-dividend period is delayed until the day after the coupon has been paid.

Japanese government bonds are generally quoted on a simple yield basis. Exceptions are JF and JGBi bonds, which are quoted on the usual clean price basis, and when-issued JGB, which are quoted in compound yield. As in many Asian countries, it is common to refer to bonds by series number, and there are separate series for all JGB types and original maturities. Although it has no natural investor base, the 10Y JGB has benchmark status[3].

[3] The (simple) 10Y yield appears on the font page of the Nikkei newspaper (Nihon Keizai Shinbun) and was also the reference point when yield curve control was introduced by the Bank of Japan.

JGB	
Maturities	2Y (JN), 5Y (JS), 10Y (JB), 20Y (JL), 30Y (JX), 40Y (JU)
Coupon	Fixed semi-annual, 0.1% decimal steps
Yield convention	Simple yield
Settlement, daycount	T+2 Tokyo, NL/365
Issuance	Auction (JS, JN, and JB monthly, JL bimonthly, JX quarterly)
Usual volume	2.1tr (JN), 5tr (JS), 6.5–7.5tr (JB), 3tr (JL), 2.1tr (JX), 2tr (JU)
Strippable	Yes

Remarks:

Bonds are reopened if they have the right maturity and a par coupon, otherwise a new bond is issued. This means in particular for 10Y (JB) issues that the final issue size depends on yield movements between the first and last auction of a new maturity point.

JF	
Maturities	15Y
Coupon	Floating rate, semi-annual
Yield convention	Not applicable
Settlement, daycount	T+2 Tokyo, NL/365
Issuance	Quarterly auction
Usual volume	1.4tr
Strippable	No

Remarks:

The coupon of a JF is reset every six months to the simple yield of the most recent 10Y auction before fees, minus a spread called alpha. The higher the alpha of a bond, the lower the future coupons, but coupons are floored at zero. When dealers bid for new JF, bids are submitted in terms of alpha. Japanese yields are very low and there is a signficant probability that a 10Y auction happens at a lower yield than the alpha of a JF, so the coupon floor of a JF usually has a meaningful value. More importantly, because the coupon is reset to a 10Y rate, JF are highly convex products and the convexity adjustment represents a large part of the value of a JF. JF issuance ended in 2008 with JF48.

JGBi	
Maturities	10Y
Coupon	Fixed semi-annual, 0.1% decimal steps, inflation-linked
Yield convention	True Street (bond-equivalent)
Settlement, daycount	T+2 Tokyo, NL/365
Issuance	Quarterly auction
Usual volume	up to 600bn
Strippable	No

Remarks:

JGBi are linked to non-seasonally adjusted Japanese CPI excluding fresh food. Unlike most other inflation-linked markets, JGBi redemptions up to JGBi16 are not floored at par. A new series of JGBi was launched in 2013 with a floor for the principal and the last coupon. In the absence of inflation, the JGBi market has been dormant since shortly after its inception in 2004.

JTB/JFB	
Maturities	13 weeks, 6 and 12 months
Coupon	Zero coupon
Yield convention	Simple yield
Settlement, daycount	T+2 Tokyo (act/365)
Issuance	Weekly auctions
Usual volume	8tr
Strippable	No

39.4 SWEDEN

The debt of the Kingdom of Sweden has been managed since 1789 by the Swedish National Debt Office (Riksgäldskontoret, now Riksgalden, www.riksgalden.se) which is one of the oldest and most sophisticated debt managers globally. Sweden issues debt in Swedish krona as well as euros and dollars and inflation-linked bonds.

The Swedish market is an important diversification market for currency speculators who are at times fairly active at the front end of the market. Like most Scandinavian countries, Sweden greatly values transparency and timely market information is widely disseminated. The Swedish market generally trades on quotes in compound yield terms.

SGB (Statsobligationer)	
Maturities	10Y, 25Y
Coupon	Annual fixed, 1/8% steps
Yield convention	True Street, annual compound
Settlement, daycount	T+2 Stockholm, ISMA 30/360
Issuance	Two auctions per month
Usual volume	40–100bn
Strippable	No

Remarks:

The euro-denominated eurobond issue SWED 5% 01/09 was fungible with the domestic SGB 5% 01/09 should Sweden adopt the euro instead of the Swedish krona.

SGBi (Realobligationer)	
Maturities	10Y, 25Y
Coupon	Annual fixed, 1/8% steps, inflation-linked
Yield convention	True street, annual compound
Settlement, daycount	T+2 Stockholm, ISMA 30/360
Issuance	Two auctions per month
Usual volume	20–60bn
Strippable	No

Remarks:

Unlike many other markets, Swedish linkers do not always have their own specific base CPI.

39.5 UNITED KINGDOM

The United Kingdom arguably has the oldest intact government bond market by virtue of having avoided invasion by another sovereign. Occasional support from the Bank of England has also been important in keeping the government solvent. Britain has a long history of circulating capital, which has been conducive to the development of capital markets in general. While it used to be quite common in other countries for private persons to store surplus wealth in cash, real assets, or bilateral loans, Britain has had a history of institutional fund management and thereby a more active allocation of capital to profitable causes. This means that the UK has set a few trends for the global bond market.

Over the centuries, Britain has created a number of bonds that are not unique to the UK, but where the UK name is applied as a generic term. The main one is consol which these days is deemed by many to be identical to the term annuity but is in fact the short form of Consolidated, which was the name given to callable perpetual bonds issued by the UK government to consolidate a number of outstanding debt securities. Another one is war loan, which in the UK refers to specific securities, but is used globally to refer to debt securities with below-market interest rates issued by governments in times of war to solicit funds from patriotically spirited citizens.

The debt of Her Brittanic Majesty The Queen's Government is managed by the Debt Management Office (DMO, www.dmo.gov.uk), which is part of Her Majesty's Treasury but operates at arm's length. Government bonds in the UK are usually called gilt-edged securities, or gilts for short. Trading in gilts is done by gilt-edged market makers (GEMMs) which are securities houses that have applied for this status at the DMO. There are no significant restrictions to being, or ceasing to be, a GEMM, but once a house resigns from GEMM status, it cannot reapply for the status within a certain grace period.

The long end of the UK bond market is driven mainly by pension funds. The UK has one of the largest funded pension systems worldwide and any change in asset allocation by pension funds, whether forced by regulation or voluntary, tends to have a large impact on the UK curve.

UKT (Treasury stock, Gilt, Conventional)

Maturities	5Y, 10Y, 20Y, 30Y, 50Y
Coupon	Fixed semi-annual, 1/8% steps
Yield convention	True street, bond-equivalent
Settlement, daycount	T+0 or T+1 London, act/act
Issuance	Auction, 1–2 per month
Usual volume	10bn
Strippable	Yes

Remarks:
A futures contract on 10Y gilts is traded on ICE London.

UKTI (Index-linked Treasury stock, index-linked Gilt)

Maturities	10Y, 20Y, 30Y
Coupon	Fixed semi-annual, 1/8% steps, inflation-linked
Yield convention	True street, bond-equivalent
Settlement, daycount	T+0 or T+1 London, act/act
Issuance	Auction, roughly monthly
Usual volume	6bn
Strippable	No

Remarks:
Index-linked Gilts are linked to UK RPI with a 3 months, earlier an 8 months lag.

39.6 UNITED STATES OF AMERICA

The US bond market is the largest in the world and the creation of this market by Alexander Hamilton was a founding moment in the history the United States [14]. US government debt is issued by the US Treasury (www.treasury.gov). The maturity points of the curve where the Treasury issues are 2Y, 3Y, 5Y, and 10Y (collectively known as Notes), and in the past, 30Y (Bonds). The expression 'the bond' therefore usually refers to the most recent 30 year issue. The US also issues T-bills and inflation-linked bonds known as TIPS.

The US operates a Primary Dealer system which is unique in that the trading of bonds is also tied to the provision of central bank liquidity (Fed Funds) into the banking system. Actual liquidity provision in the secondary market has, however, spread across multiple firms thanks to easy access to anonymous matching platforms and, more recently, central clearing. The US bond market relies to a lesser extent than the European and Japanese market on bond futures contracts for price discovery although bond futures volumes are now considerable. The main price discovery takes place through the on-the-run benchmark securities. The US market has so far resisted decimalisation.

Tentative auction calendars are available some time ahead, but the actual size of supply generally becomes known through the Quarterly Refunding Announcement.

T (Treasury note/Treasury bond)	
Maturities	2Y, 3Y, 5Y, 10Y, 30Y
Coupon	Fixed semi-annual, currently 1/8% increments, decimal considered
Yield convention	True street, bond-equivalent
Settlement, daycount	T+1 New York (different for auctions), act/act
Issuance	Several auctions per month
Usual volume	40-50 bn
Strippable	Yes

TIPS/TII (Treasury Inflation-protected security)	
Maturities	5Y, 10Y, 30Y
Coupon	Fixed semi-annual, 1/8% steps
Yield convention	True street, bond-equivalent
Settlement, daycount	T+1 New York (different for auctions)
Issuance	Quarterly auction
Usual volume	15–40bn
Strippable	Yes

B (Treasury bill)	
Maturities	4, 13, 26, 51 weeks
Coupon	Zero coupon
Yield convention	Discount yield
Settlement, daycount	T+1 New York, act/360
Issuance	Weekly auction (4 week bill Mondays, 13 and 26 weeks Thursday
Usual volume	30–120bn
Strippable	No

Remarks:

The close spacing of bills makes them bellwether instruments for market sentiment regarding timeliness of payment by the US Treasury during periods of shutdowns. These are times when Congress has not approved the bills required to fund the US government expenditure and the market therefore prices in the risk of temporary defaults.